

Game theory and the prisoners' dilemma

The branch of economics that considers strategic decision-making is known as game theory, and the prisoners' dilemma is perhaps its best-known example. Two thieves are captured, questioned separately and urged to give evidence against each other. They know that if both stay silent they will receive a light sentence of a year in prison. Alternatively, if one of them betrays their partner and the other stays silent, the former is released and the latter will be sentenced to 20 years. If both betray, however, they will each get 10 years. Since the worst outcome is clearly to get double-crossed, they both end up betraying each other and receive 10 years each. But by acting rationally the thieves have missed out on the best outcome – that of mutual silence. An economic example of a prisoners' dilemma is a cartel – a group of firms that agree to restrict output to keep prices high. Given high prices, it is rational for a firm to produce more to reap large additional profits, but when all firms do this, output shoots up and prices collapse, undermining the cartel's original aim.

		Prisoner 1	
		Confess	Keep silent
Prisoner 2	Confess	Both jailed for 10 years.	Prisoner 1 jailed for 20 years. Prisoner 2 walks free.
	Keep silent	Prisoner 1 walks free. Prisoner 2 jailed for 20 years.	Both jailed for 1 year.